UNIVERSITÀ DEGLI STUDI DI UDINE
hic sunt futura

DEPARTMENT OF MATHEMATICS, COMPUTER SCIENCE AND PHYSICS

Master Degree: Artificial Intelligence & Cybersecurity

A comparative analysis of Latent Architectures

Structural Optimization and Discriminative Power in High-Dimensional Data.

Damiano Fumagalli **ID: 157547**

Contact: fumagalli.damiano@spes.uniud.it

2025/2026

Manifold Hypothesis: in high-dimensional spaces, data becomes sparse, and Euclidean distances become uninformative. This sparsity implies that the data occupies a lower-dimensional Manifold**.**

Goal of the presentation: show how different autoencoder architectures exploit this manifold for feature extraction and data compression

Case Study: using Fashion-MNIST dataset train different kind of autoencoders and evaluate the latent space fidelity with some metrics and classification tasks

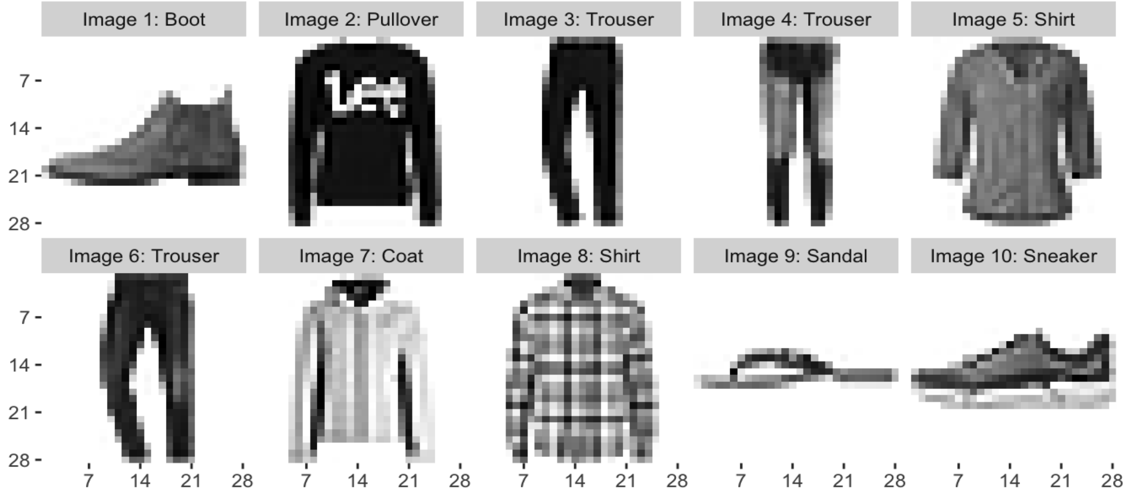


Figure 1: Samples from Fashion-MNIST dataset

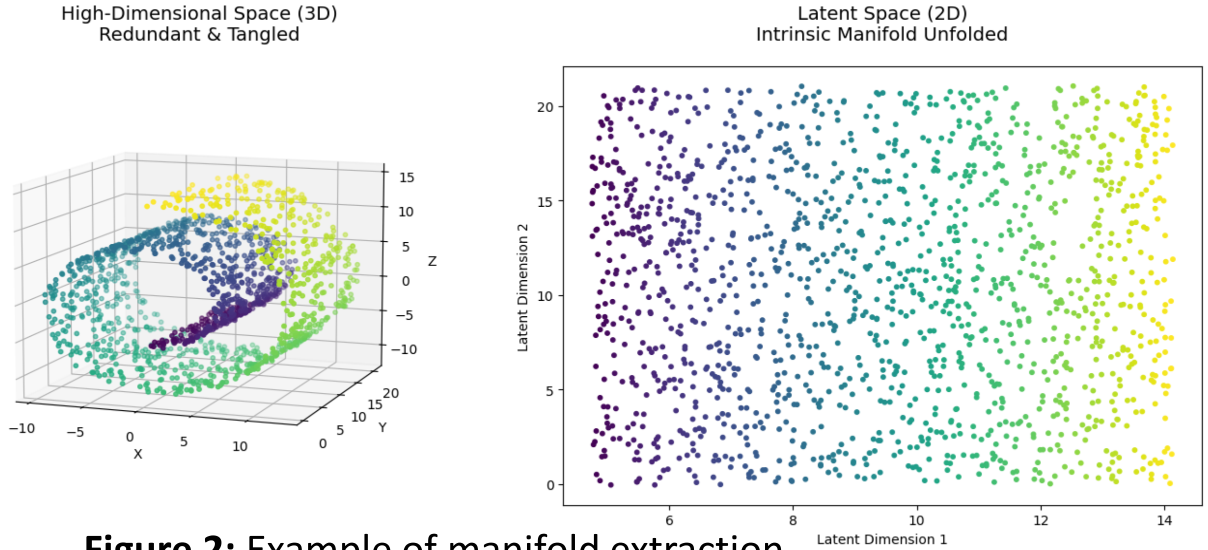
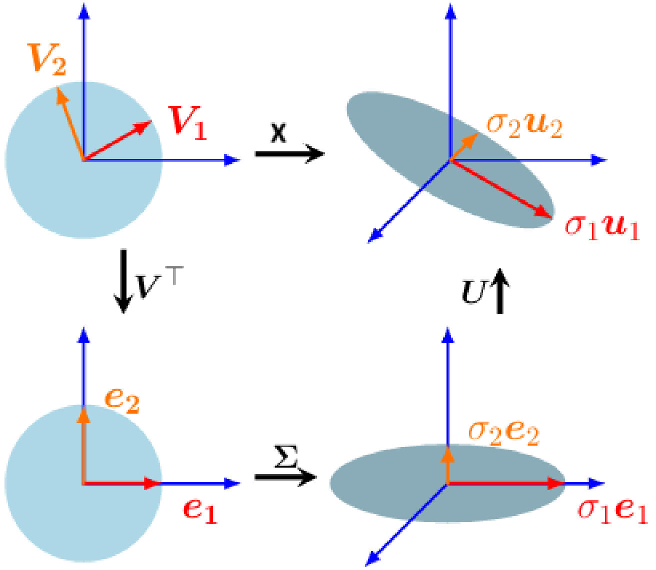
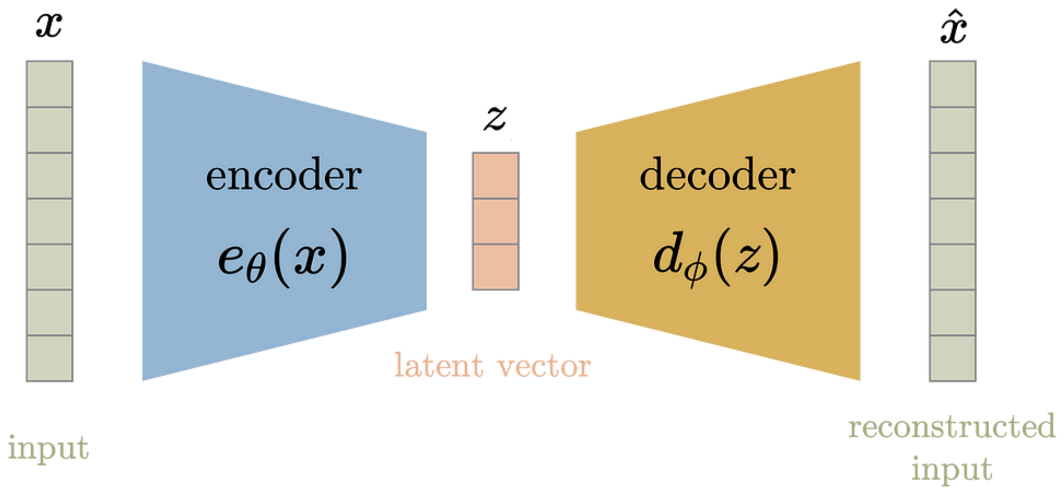


Figure 2: Example of manifold extraction



- $X \in R^D \rightarrow Z \in R^d, d \ll D$
- **PCA (k):** $Z = XV_k$
  - Goal: minimize the reconstruction loss $L = \left|\left|X - XV_kV_k^T\right|\right|_F^2$
  - V_k is obtained by the first k components of the right singluar values V of the SVD
  - **SVD:** $\widehat{X} = U\Sigma V^T$



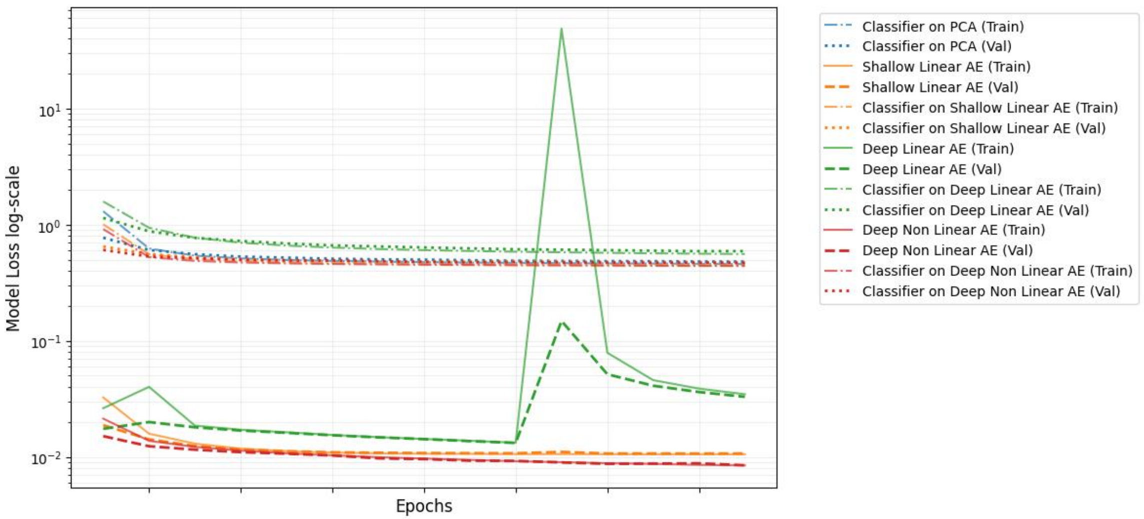
$$loss = \|x - \hat{x}\|_2 = \|x - d_\phi(z)\|_2 = \|x - d_\phi(e_\theta(x))\|_2$$

- $e_\theta(x): R^D \rightarrow R^d$ produces $Z = XW_\theta$
- $d_\phi: R^d \rightarrow R^D$ produces $\hat{X} = ZW_\phi = XW_\theta W_\phi$
- Goal: minimize reconstruction loss $L = \left|\left|X - XW_\theta W_\phi\right|\right|_F^2$

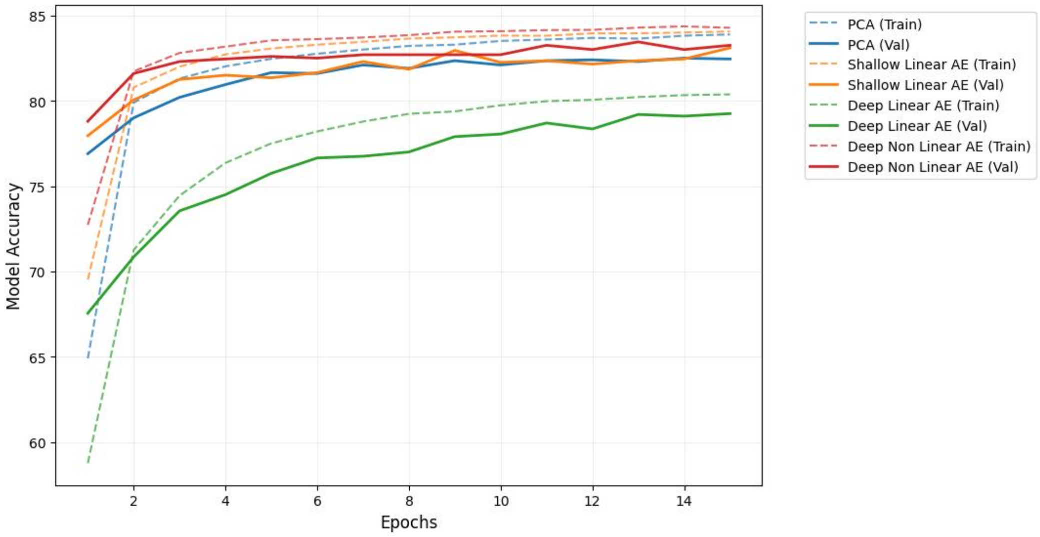
UNIVERSITÀ DEGLI STUDI DI UDINE
hic sunt futura

DEPARTMENT OF MATHEMATICS, COMPUTER SCIENCE AND PHYSICS

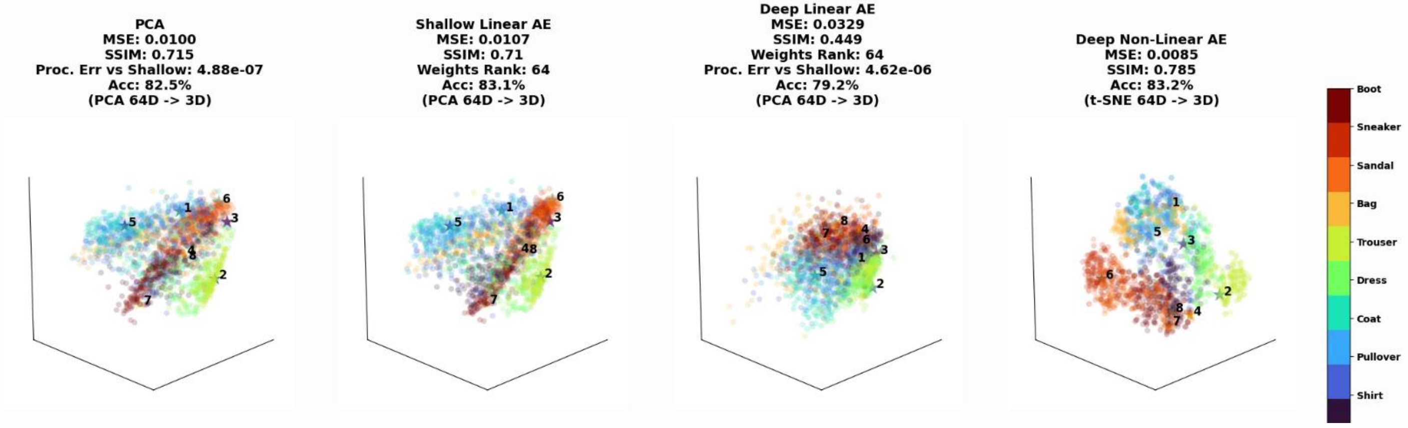
Training & Validation Loss



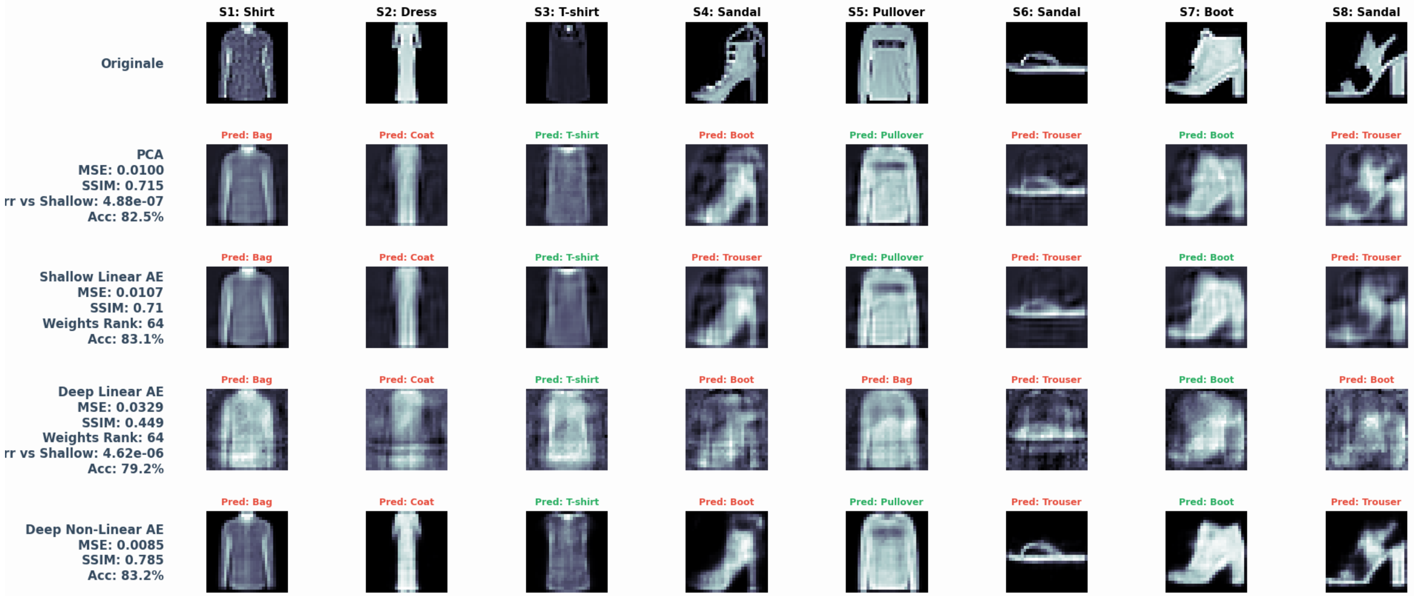
Training & Validation Accuracy

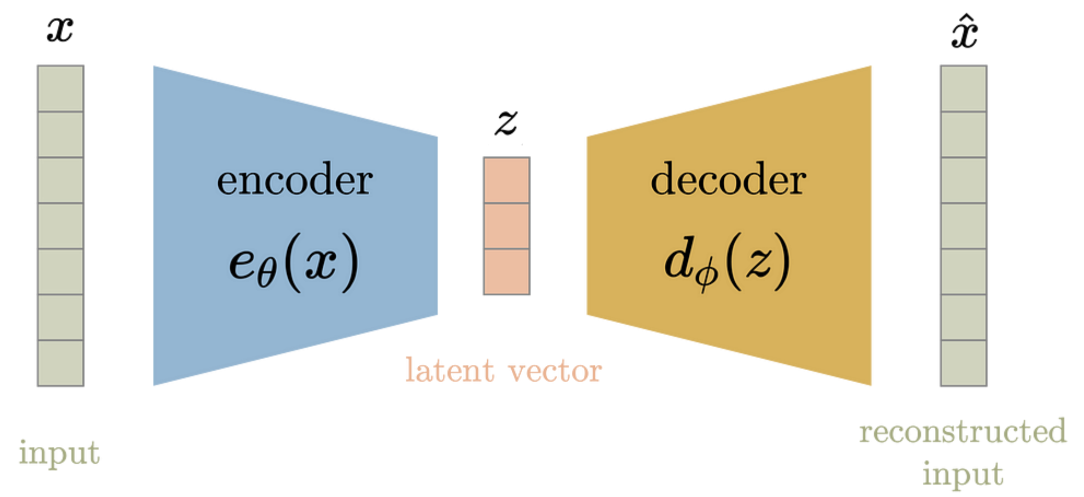


64D Latent Space Comparison

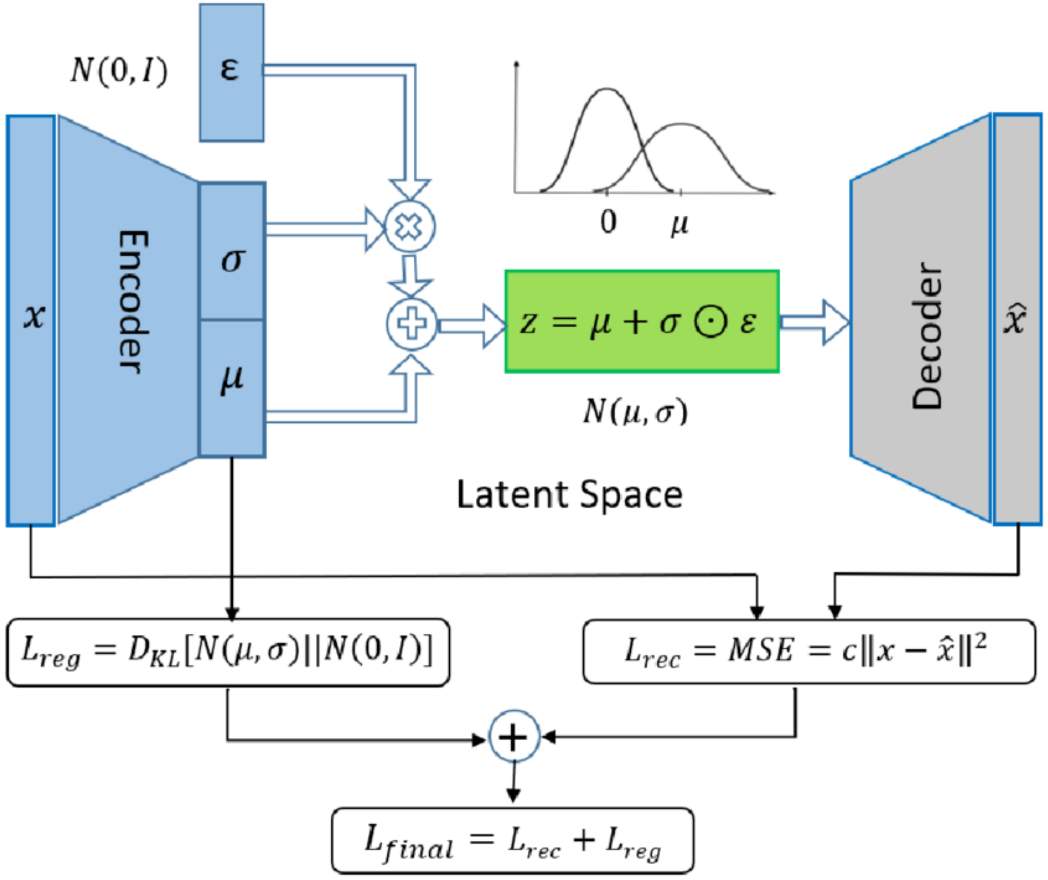


Sample Reconstruction Comparison



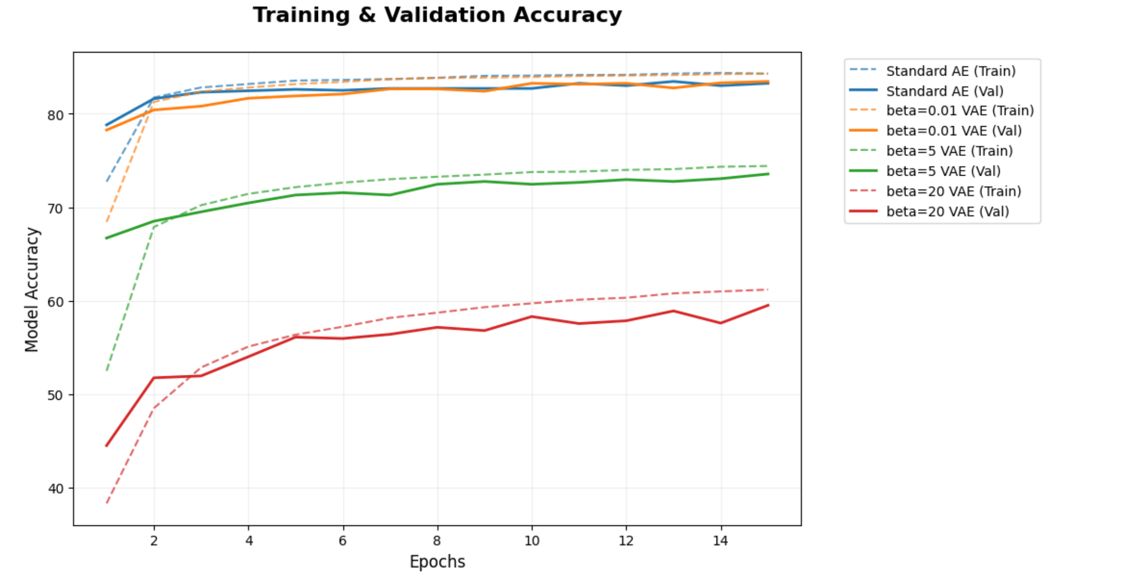


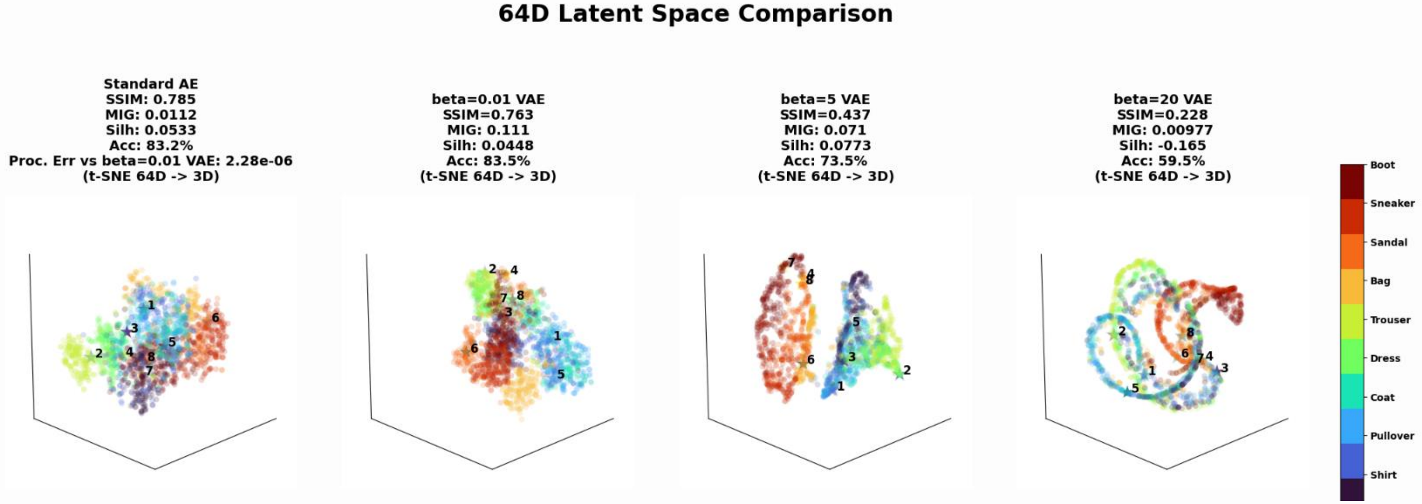
$$loss = \|x - \hat{x}\|_2 = \|x - d_\phi(z)\|_2 = \|x - d_\phi(e_\theta(x))\|_2$$

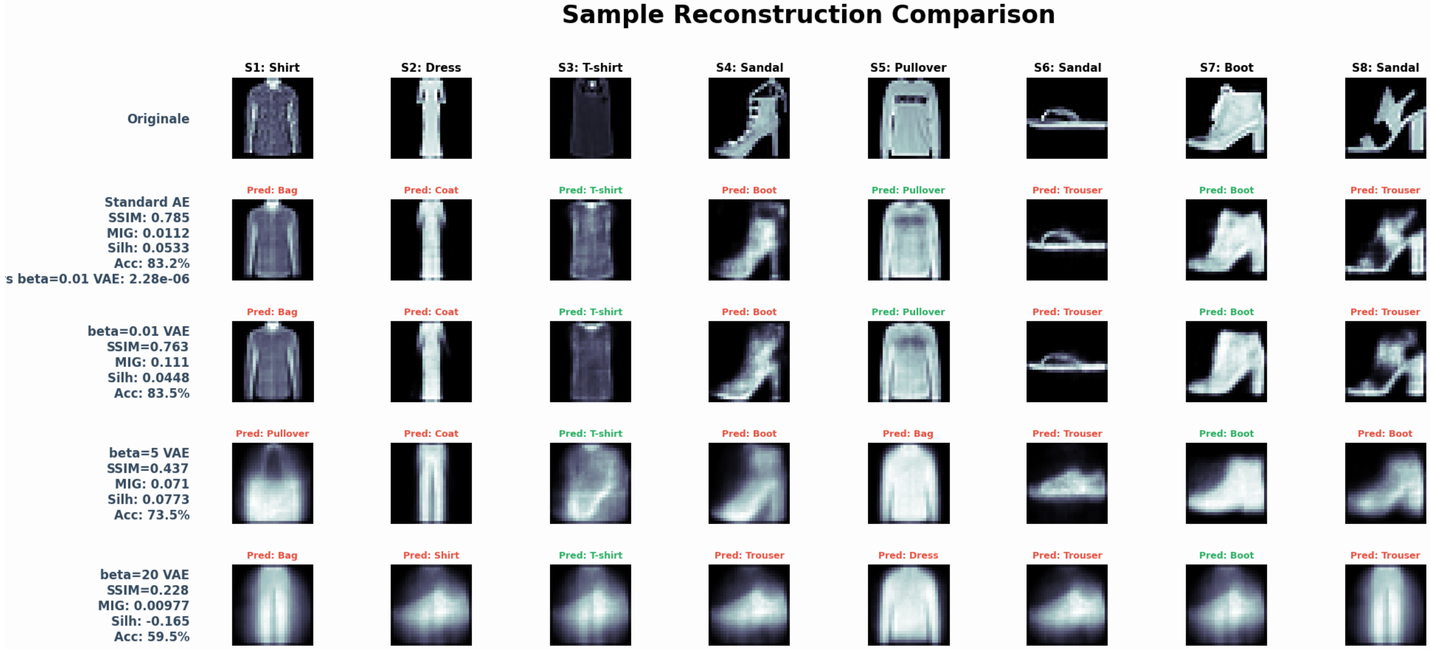


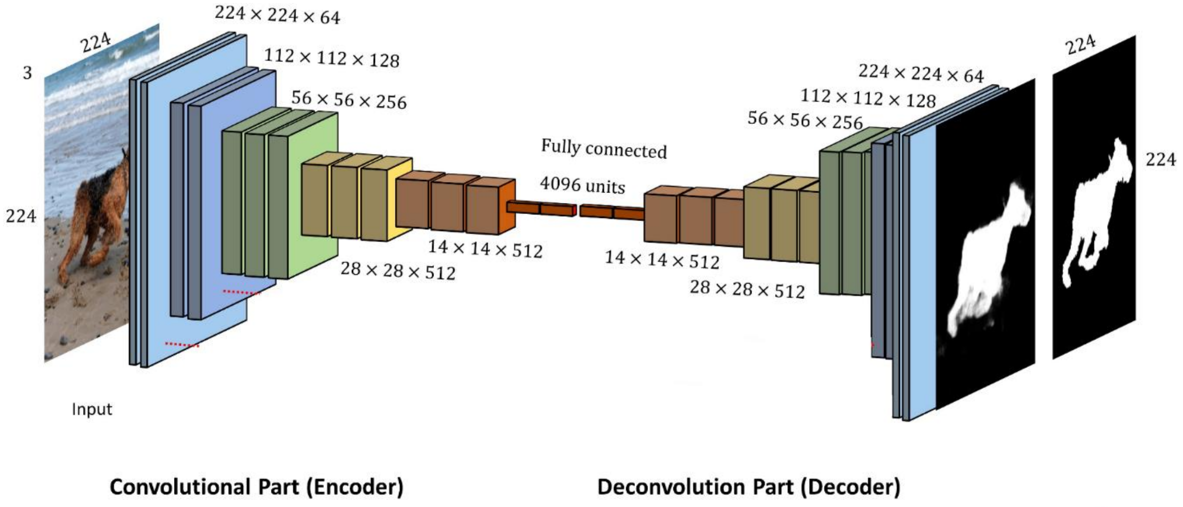
UNIVERSITÀ DEGLI STUDI DI UDINE
hic sunt futura

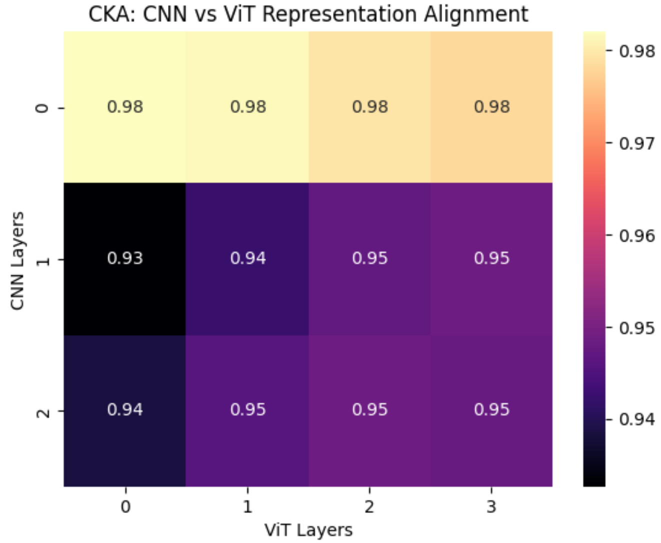
DEPARTMENT OF MATHEMATICS, COMPUTER SCIENCE AND PHYSICS

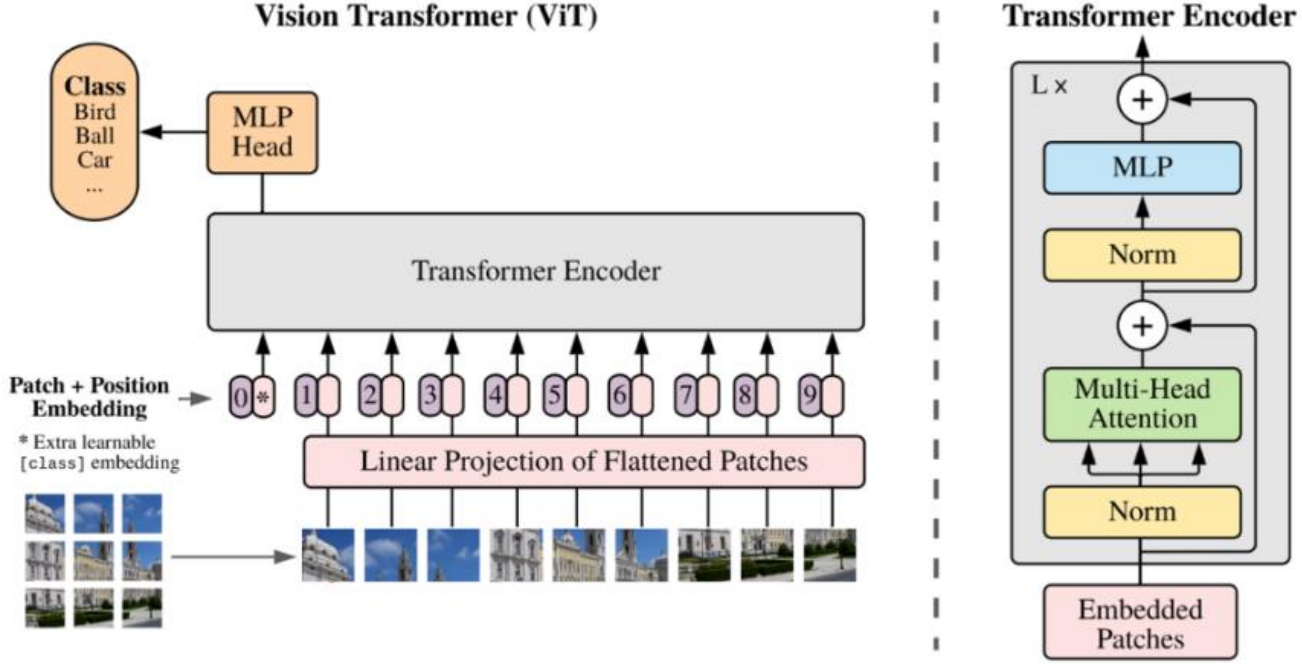












- [“Do Vision Transformers See Like Convolutional Neural Networks?”]
  Maithra Raghu, Thomas Unterthiner
- [“On the Relationship between Self-Attention and Convolutional Layers”]
  Jean-Baptiste Cordonnier, Andreas Loukas, Martin Jaggi

UNIVERSITÀ DEGLI STUDI DI UDINE
hic sunt futura

DEPARTMENT OF MATHEMATICS, COMPUTER SCIENCE AND PHYSICS